

## Ejercicio 1b

Simon Fonseca

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In the rest frame of one B-meson:

$$p_1 = (m_B, \vec{0}), \quad p_2 = \left( \frac{m_B}{\sqrt{1-v_{rel}^2}}, \frac{m_B v_{rel}}{\sqrt{1-v_{rel}^2}} \right)$$

The scalar product is:

$$p_1 \cdot p_2 = \frac{m_B^2}{\sqrt{1-v_{rel}^2}}$$

But this must equal the same scalar product in any frame. In the lab frame:

$$p_1 \cdot p_2 = \mathcal{E}_1 \mathcal{E}_2 - \vec{p}_1 \cdot \vec{p}_2$$

We want to find  $p_1 \cdot p_2 = p_{B^+} \cdot p_{B^-}$  in the lab frame. 4-momentum is conserved:

$$p_{e^-} + p_{e^+} = p_{B^+} + p_{B^-}$$

Computing  $(p_{B^+} + p_{B^-})^2$ :

$$(p_{B^+} + p_{B^-})^2 = p_{B^+}^2 + p_{B^-}^2 + 2p_{B^+} \cdot p_{B^-}$$

Since  $p_{B^+}^2 = m_B^2$  and  $p_{B^-}^2 = m_B^2$  (mass-shell condition):

$$(p_{B^+} + p_{B^-})^2 = m_B^2 + m_B^2 + 2p_{B^+} \cdot p_{B^-} = 2m_B^2 + 2p_{B^+} \cdot p_{B^-}$$

But from conservation,  $(p_{B^+} + p_{B^-})^2 = (p_{e^-} + p_{e^+})^2 = s$ :

$$s = 2m_B^2 + 2p_{B^+} \cdot p_{B^-}$$

Solving for the dot product:

$$p_{B^+} \cdot p_{B^-} = \frac{s - 2m_B^2}{2}$$

For massless electrons/positrons moving in opposite directions along z:

$$p_{e^-} = (E_1, 0, 0, E_1), \quad p_{e^+} = (E_2, 0, 0, -E_2)$$

Adding them:

$$p_{e^-} + p_{e^+} = (E_1 + E_2, 0, 0, E_1 - E_2)$$

For a 4-vector  $p^\mu = (E, p_x, p_y, p_z)$ , the invariant square is:

$$p^2 = E^2 - p_x^2 - p_y^2 - p_z^2$$

So:

$$s = (p_{e^-} + p_{e^+})^2 = (E_1 + E_2)^2 - 0^2 - 0^2 - (E_1 - E_2)^2$$

$$s = (E_1 + E_2)^2 - (E_1 - E_2)^2$$

Expanding:

$$(E_1 + E_2)^2 = E_1^2 + 2E_1E_2 + E_2^2$$

$$(E_1 - E_2)^2 = E_1^2 - 2E_1E_2 + E_2^2$$

Subtracting:

$$s = (E_1^2 + 2E_1E_2 + E_2^2) - (E_1^2 - 2E_1E_2 + E_2^2) = 4E_1E_2$$

Finally:

$$p_{B^+} \cdot p_{B^-} = \frac{4E_1E_2 - 2m_B^2}{2} = 2E_1E_2 - m_B^2$$

Therefore:

$$\frac{m_B^2}{\sqrt{1 - v_{rel}^2}} = 2E_1E_2 - m_B^2$$

Solving for  $v_{rel}$ :

$$v_{rel} = \sqrt{1 - \frac{m_B^4}{(2E_1E_2 - m_B^2)^2}}$$